

Pharmacist Integration to Support Continuous Glucose Monitoring Initiation: A Collaborative, Patient-Centered Approach

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James Thurston, Pharm.D¹ , Hanlin Li, PharmD, MBA¹ , Mangala Rajan, MBA², Yuliya Baratt, PharmD, MBA¹, Amber Bradley, PharmD¹, and Fred Pelzman, MD²

Abstract

Objective: The development of continuous glucose monitoring (CGM) has allowed for improved glycemic control among patients with diabetes. Clinical pharmacists possess medication expertise and can provide support for increased CGM utilization through device education and affordability assistance, but there is limited evidence evaluating the effectiveness of clinical pharmacist-assisted CGM initiation. The objective of this study was to examine how clinical pharmacist-assisted CGM implementation can impact glycemic control for patients with diabetes. **Methods:** This is a retrospective pre-post study that evaluated change in A1c among patients who were assisted with CGM device implementation by a clinical pharmacist between January 1, 2019, and December 31, 2023. The primary outcome of this study was change in A1c from baseline (prior to CGM initiation) to the next subsequent A1c following CGM initiation. The study team also investigated change in A1c among a subgroup of patients followed independently by clinical pharmacists practicing under a collaborative drug therapy management (CDTM) agreement. **Results:** Pharmacist-assisted CGM initiation led to a statically significant decrease in mean A1c of -0.71 (CI 95% $0.41-1.00$, $P < 0.001$) across all patients. Within the CDTM subgroup, the mean A1c difference was -1.60 (CI 95% $0.64-2.55$, $P = 0.002$) while in the non-CDTM subgroup, the mean A1c difference was -0.50 (CI 95% $0.22-0.78$, $P < 0.001$). **Conclusions:** Clinical pharmacists are effective at helping patients with diabetes reduce their A1c through assisting with CGM initiation, education, and follow-up. Among patients included in this study, those followed by pharmacists practicing under CDTM agreements saw the greatest amount of A1c reduction.

Keywords

pharmacists, continuous glucose monitoring, diabetes mellitus

Introduction

Diabetes mellitus is a chronic, complex disease state that requires regular monitoring to ensure appropriate glycemic control, while avoiding adverse effects, such as hypoglycemia. In the past, glucose monitoring techniques were limited to traditional blood glucose monitors (BGM) through fingerstick tests, but more recently, continuous glucose monitoring (CGM) technology has shifted the landscape of diabetes management. CGM devices measure interstitial glucose levels continuously over time, allowing for a broader view of glycemic control throughout the day. They are placed on or under the skin, where they remain attached for an extended period, thus limiting the burden of regular fingerstick measurements with traditional BGM. Some devices also have unique features that improve their utility, such as hypo- and hyperglycemia alerts, prediction of glucose trends, and data-sharing capabilities. CGM devices can be classified as either professional

CGMs, which are clinic-based devices intended for short-term use (7-14 days), or personal CGMs, which are applied by the patient at home and intended for long-term use.¹

In addition to improved ease of monitoring over traditional BGM, CGM has also been shown to improve patients' glycemic control. The American Diabetes Association (ADA) has recommended that CGM should be offered for all patients with Type 1 or Type 2 diabetes who use basal insulin, insulin

¹Department of Pharmacy, New York-Presbyterian Hospital, New York, NY, USA

²Department of Medicine, New York-Presbyterian Hospital/Weill Cornell Medical Center, New York, NY, USA

Corresponding Author:

James Thurston, PharmD, Department of Pharmacy, New York-Presbyterian Hospital, 622 West 168th Street, New York, NY 10032, USA.

Email: jthurst6@gmail.com

pumps, or multiple daily injections of insulin.² This recommendation is based on numerous randomized controlled trials that have demonstrated that CGM use can reduce A1c and/or hypoglycemia vs BGM among patients with both Type 1 and Type 2 diabetes.³⁻⁷ As CGM technology has continued to develop and expand, a greater number of patients have been able to access these beneficial monitoring devices.^{8,9}

Despite the benefits of CGM, certain barriers exist that limit patients' ability to start CGM, including insufficient device education and concerns regarding insurance coverage and cost. Within the health care team, clinical pharmacists have unique medication and device expertise and can assist with overcoming these barriers. Clinical pharmacists working in outpatient settings can help with a variety of services that support CGM implementation such as device education, CGM data analysis, and prior authorization / access assistance. Additionally, within New York State, eligible pharmacists can create collaborative drug therapy management (CDTM) agreements with physicians, allowing pharmacists to independently adjust and manage patients' medication therapy to optimize care in addition to their other routine appointments.¹⁰ Through CDTM agreements, clinical pharmacists can utilize CGM data to optimize medication use and improve clinical outcomes for those with diabetes.

While recent literature has shown that pharmacists can improve patients' glycemic control through medication management, there is a need for more literature evaluating clinical pharmacists' role in improving patients' glycemic control through initiation of CGM devices.¹¹ Of prior studies that have specifically examined the effect of clinical pharmacists facilitating CGM initiation, available studies either did not evaluate clinical outcome or evaluated the use of professional CGMs, which are not intended for long-term use.^{12,13} The current gap in literature necessitated this study evaluating the glycemic effects of clinical pharmacist-assisted personal CGM initiation.

Methods

The specific aim of this study was to identify if clinical pharmacist assistance with CGM implementation is beneficial for glycemic improvement for patients with diabetes. Furthermore, this study sought to identify what services pharmacists may provide for these patients under a CDTM agreement, and how this increased autonomy affects A1c reduction.

Setting

For this retrospective review, we evaluated CGM implementation-related interventions by clinical pharmacists in one internal medicine clinic and one endocrinology specialty clinic at an academic medical center in New York City. Providers in these practices referred patients to the clinical pharmacist for independent CGM initiation services, where clinical pharmacists assessed CGM appropriateness, provided

device education, and facilitated CGM accessibility through insurance approval assistance. Some patients, in addition to these services, had follow-up visits with the clinical pharmacist, which included confirmation of device receipt, discussion of appropriate techniques for CGM use, and education on how to access or share CGM data. Following CGM initiation, patients would either continue seeing their primary care provider (PCP) or endocrinologist for usual care or see a clinical pharmacist at independent CDTM visits in addition to their usual care. This diabetes CDTM agreement has been previously established within the internal medicine clinic and permits clinical pharmacists to independently optimize medications during these subsequent patient visits.

Study Design

This was a single-center, retrospective pre-post study that evaluated change in A1c among all patients who were assisted with CGM device implementation by a clinical pharmacist between January 1, 2019, and December 31, 2023. Retrospective chart review included patient demographics, A1c measurements, clinical pharmacist interventions, and patients' reasons for CGM discontinuation or refusal. This protocol was reviewed and approved by the Weill Cornell Medicine Investigational Review Board (IRB) and determined to qualify for exemption and for waivers of informed consent and HIPAA authorization.

Patients were included in this review if they were at least 18 years of age with a diagnosis of diabetes mellitus or latent autoimmune diabetes in adults (LADA). Patients were excluded if they were unable to initiate the CGM device, if a professional CGM device was used, or if they did not have A1c data or clinical follow-up at the study sites within one year of CGM initiation. Patients were also excluded if it was explicitly stated within chart documentation that the patient discontinued their CGM device prior to the repeat A1c measurement.

The primary outcome of this study was change in A1c from baseline to the next A1c following CGM initiation. Baseline A1c was defined as the most recent A1c measurement taken prior to CGM initiation. Repeat A1c was defined as the next subsequent A1c measurement taken at least three months after CGM initiation.

A secondary outcome of this study was change in A1c specifically among a subgroup of patients who were followed independently by clinical pharmacists under the diabetes CDTM agreement in addition to usual care. Other secondary outcomes included quantification of the types of clinical pharmacist interventions performed.

This study methodology was developed using guidance provided by Bauman et al¹⁴ to ensure high-quality, methodological rigor for this retrospective review.

Statistical Analysis

We profiled the change in A1c and determined if the distribution was close to normal. We then analyzed each variable

with change in A1c to determine if any of the relationships were significant. We used the appropriate parametric (correlation) or non-parametric tests (Kruskal-Wallis), depending on the nature of the variables. Based on the analysis, we planned on selecting variables with a significance of less than 0.1 for entry into a regression model.

This study was constrained by the available patient pool, which dictated the sample size used, and hence, power analysis was not considered. We view this study as a precursor for larger studies for which sample size and power may be more pertinent considerations. R statistical software was utilized for all statistical analyses.

Results

Of the 131 eligible patients, 110 were included for study analysis, and 21 patients were excluded due to professional (instead of personal) CGM use, lack of insurance coverage, or failure to obtain a repeat A1c within one year of CGM initiation for any reason. Details are listed in Figure 1.

All baseline characteristics and demographic data are reported in Table 1. Compared with the internal medicine practice site, patients referred from the endocrinology clinic were younger (mean age of 48.1 vs 59.2) and more often had Type 1 diabetes (46% vs 19.2%). Endocrinology clinic patients also more often reported being White (20.6% vs 10.6%) and less often reported being Asian (1.6% vs 12.8%). Prior to CGM initiation, >96% of patients in both groups were on insulin, although insulin pumps were more common among patients referred from the endocrinology practice (9.5% vs 2.1%). The three patients who were not on insulin at baseline each had Type 2 diabetes and either had previously been on insulin or were on multiple other medications for diabetes.

Patient characteristics were similar regarding private insurance status (14.3% vs 12.8%) and preferred language.

Primary Endpoint

At baseline, patients referred to a clinical pharmacist for CGM initiation had a mean A1c of 8.7% (SD 1.96). After at least three months after CGM initiation by a clinical pharmacist, mean A1c decreased to 8.0% (SD 1.69), representing a significant decrease (mean difference -0.71 , CI 95% 0.41 - 1.00 , $P < 0.001$). Change in A1c among patients referred from each clinic as well as change in A1c among all patients is represented in Table 2.

The baseline characteristics of age, race, sex, diabetes status and insulin use were individually analyzed for association with our primary outcome, change in A1c. The association between these descriptive variables and change in A1c are represented in Table 3, which shows that no variables were statistically significant at the threshold of $p \leq 0.1$. As a result, we did not perform a regression as the next step.

Secondary Endpoints

A subgroup analysis is summarized in Table 4. Patients followed by usual care ($n = 89$) after CGM initiation had a mean baseline A1c of 8.3% (SD 1.64). After at least three months of CGM use, this mean A1c decreased to 7.8% (SD 1.64), representing a significant change (mean difference -0.50 , CI 95% 0.22 - 0.78 , $P < 0.001$). Alternatively, patients managed by a clinical pharmacist practicing under CDTM ($n = 21$) had a mean baseline A1c of 10.3% (SD 2.39). After at least three months of CGM use, this mean A1c decreased to 8.7% (SD 1.78), representing an even greater magnitude of reduction in

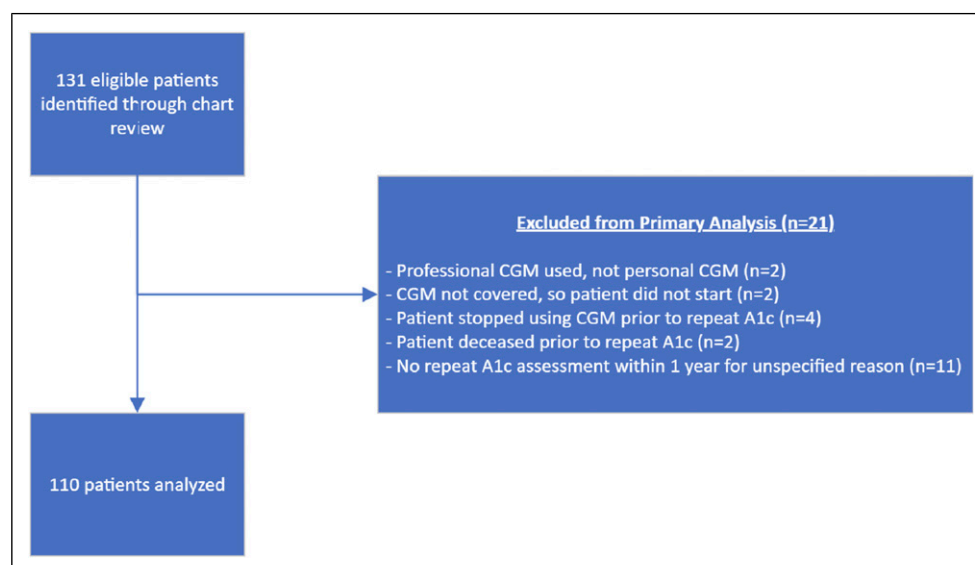


Figure 1. Participant Enrollment.

Table 1. Baseline Patient Characteristics.

	Endocrinology Clinic (n = 63)	Internal Medicine Clinic (n = 47)
Age at time of CGM initiation, mean, (SD)	48.1 (15.4)	59.2 (15.2)
Sex, Male – no. (%)	21 (33.3%)	21 (44.7%)
Race – no. (%)		
Black or African American	22 (34.9%)	14 (29.8%)
White	13 (20.6%)	5 (10.6%)
Asian	1 (1.6%)	6 (12.8%)
American Indian or Alaska Nation	1 (1.6%)	1 (2.1%)
Other combinations	22 (34.9%)	18 (38.3%)
Unknown / Declined	4 (6.4%)	3 (6.4%)
Ethnicity – no. (%)		
Hispanic or Latino	23 (36.5%)	11 (23.4%)
Not Hispanic or Latino	36 (57.1%)	30 (63.8%)
Unknown / Declined	4 (6.4%)	6 (12.8%)
Preferred language – no. (%)		
English	54 (85.7%)	43 (91.4%)
Spanish	4 (6.4%)	2 (4.3%)
Other	5 (7.9%)	2 (4.3%)
Insurance – no. (%)		
Medicare	6 (9.5%)	13 (27.7%)
Medicaid	33 (52.4%)	12 (25.5%)
Medicare and Medicaid	15 (23.8%)	15 (31.9%)
Private / Commercial	9 (14.3%)	6 (12.8%)
Uninsured	0	1 (2.1%)
Type of diabetes		
Type 1	29 (46.0%)	9 (19.2%)
Type 2	33 (52.4%)	37 (78.7%)
LADA ^a	1 (1.6%)	1 (2.1%)
Overall insulin use at baseline – no. (%)	61 (96.8%)	46 (97.9%)
Type of insulin use – no. (%)		
Basal + Bolus	48 (76.2%)	41 (87.3%)
Insulin pump (basal + Bolus)	6 (9.5%)	1 (2.1%)
Basal Only	6 (9.5%)	3 (6.4%)
Bolus Only	1 (1.6%)	1 (2.1%)
None	2 (3.2%)	1 (2.1%)

^aLatent autoimmune diabetes in adults (LADA).

Table 2. Change in A1c.

	Endocrinology Clinic (n = 63)	Internal Medicine Clinic (n = 47)	All Patients (n = 110)
Baseline A1c, mean (SD)	8.8% (1.94)	8.6% (1.99)	8.7% (1.96)
Repeat A1c within one year, mean (SD)	7.9% (1.73)	8.1% (1.65)	8.0% (1.69)
Mean difference in A1c	f –0.87	f –0.49	f –0.71
P-value	P < 0.001, CI 0.50-1.24	P = 0.0445, CI 0.01-0.97	P < 0.001, CI 0.41-1.00

A1c (mean difference –1.60, CI 95% 0.64-2.55, $P = 0.002$). (See [Table 4](#)).

Among the 110 patients prior to exclusion who started a personal CGM device, all 110 (100%) were approved through insurance and the clinical pharmacist assisted all patients (100%) with CGM initiation. Of note, 104 patients were completely new to CGM, while six had tried CGM sometime

in the past but were not using this at baseline. All six of these patients were assisted with re-initiation of their CGM or assistance with insurance prior authorization/paperwork re-submission, while 7 of the patients naïve to CGM use were also assisted with this at follow-up visits. [Supplemental Table S1](#) describes the CGM-initiation interventions for all patients and [Supplemental Table S2](#) summarizes the subgroup of

Table 3. Association of Change in A1c With Descriptive Variables From Baseline.

Explanatory Variable	Test of Association	Estimate of Association	Significance
Age	Correlation	−0.02	0.8
Sex	Kruskal Wallis (KW) Chi-square	1.25	0.5
Race	KW Chi-square	1.81	0.9
Type of insulin used at baseline	KW Chi-square	5.51	0.2
Type of diabetes	KW Chi-square	1.07	0.5

Table 4. Change in A1c - Sub-Group Analysis.

	Managed by MD/NP (n = 89)	Managed by CDTM (n = 21)
Baseline A1c, mean (SD)	8.3% (1.64)	10.3% (2.39)
Repeat A1c within one year, mean	7.8% (1.64)	8.7% (1.78)
Mean difference in A1c	−0.50	−1.60
P-value	P < 0.001, CI 0.22-0.78	P = 0.002, CI 0.64-2.55

patients followed by a clinical pharmacist practicing under CDTM. Common interventions over the study period included device troubleshooting (66.7%), medication refills (42.9%), and medication dose modification (19.0%). Device troubleshooting was classified as any situations in which a clinical pharmacist assisted patients with CGM devices not appropriately collecting or transmitting glucose data, such as issues with physical device placement or failure of device to capture glucose readings. Further, the lack of association of age, race, sex, diabetes status and insulin with change in A1c show that this reduction was seen across all patients, with no one group benefitting more or less than another.

Discussion

This retrospective review demonstrated that clinical pharmacist assistance with CGM initiation may promote A1c reduction and CGM accessibility among patients with diabetes. To our knowledge, this is a novel study examining the clinical outcomes (ie, A1c) among patients who used a personal CGM device in a population of diverse racial and socioeconomic backgrounds with diabetes.

Among the 110 patients included in this study, mean A1c significantly decreased from 8.7% to 8.0% after clinical pharmacist-assisted CGM initiation, representing improvement in patient's glycemic measures. Of note, 60/110 (54.5%) of these patients had an A1c reduction of ≥ 0.5 , representing a clinically relevant reduction in disease control.^{15,16} This reduction in A1c highlights the benefits of CGM initiation, which was facilitated by clinical pharmacists in this study, allowing for improved access to the health care team, and improved timeliness of coordinating CGM initiation and teaching patients how to utilize CGM devices.

Within our subgroup analysis of patients followed by either usual care or a CDTM pharmacist after CGM initiation, our analysis demonstrated that patients from both subgroups

experienced benefits in A1c reduction. Of note, patients within the CDTM subgroup had a higher mean baseline A1c than the usual care group (mean 10.3% vs 8.3%). This may suggest that clinicians were referring patients to the clinical pharmacist who were more complex and required closer follow-up, such as patients with a history of medication intolerance, large glycemic excursions, and/or worse glycemic control. Similar findings have been observed in literature that compare pharmacist and non-pharmacist groups, validating the benefit of pharmacist practicing with a scope of practice to closely follow patients with greater clinical complexity.¹⁷ These findings also suggest that individualized sessions with a clinical pharmacist focused on device education and accessibility facilitate CGM initiation, even if a patient is not subsequently followed by a clinical pharmacist practicing under a CDTM agreement.

Despite having a higher mean baseline A1c, patients in the CDTM sub-group also experienced a greater decrease in mean A1c (mean difference −1.6 vs −0.5) than the non-CDTM sub-group. Among patients in the CDTM group, 15/21 (71.4%) had a significantly relevant A1c reduction of ≥ 0.5 , compared to 45/89 (50.6%) of the usual care group.^{15,16} Potential reasons for these differences in A1c reduction may include an increased A1c among this group at baseline, an increased frequency of CDTM visits vs usual care, and the impact of CGM-focused and medication-focused interventions conducted by clinical pharmacists at these visits. These CDTM-specific interventions, such as medication initiation and modification, likely enhanced glycemic control among patients with CGM devices. The results suggest that clinical pharmacist assistance with CGM initiation is beneficial on its own, but clinical pharmacist care under CDTM can be particularly beneficial for patients with greater clinical complexity, such as those with a history of medication intolerance, large glycemic excursions, and/or worse glycemic control.

Among all patients, even those excluded from this study, pharmacist-assisted CGM initiation resulted in 98.4% insurance approval for CGM devices. This demonstrates the importance of clinical pharmacist integration early on within the CGM initiation process to assist with prior authorization and access at the pharmacy. Among patients followed by clinical pharmacists practicing under CDTM, pharmacists assisted with several additional services such as device troubleshooting, medication refills, and medication optimization. These clinical pharmacist-led, diabetes-focused visits at our study site were more frequent than usual care visits to enhance care and allow for the provision of these services under CDTM.

This study was limited by retrospective design, which prevented our ability to schedule routine A1c measurements for patients, leading to patient exclusion and variable timing for A1c measurement after three months. Although generally taken every three months for patients with uncontrolled diabetes, this variable timing of A1c measurement may have impacted the assessment of change in A1c among this population. Of note, the timeframe of this study included the COVID-19 pandemic, when access to clinic appointments for labs was not routinely available. This variable A1c timing mirrors real-life scenarios where not all patients are able to obtain A1c in a timely manner and is similar to the design of other pharmacist-focused interventional trials that assess change in A1c.¹⁸⁻²⁰ Future similar studies may find benefit in assessing the progression of A1c over a longer timeframe (ie, 1 year) and evaluating the time to reduction in A1c.

The retrospective nature of this study also limited our data collection to chart review, which may not have captured any undocumented interventions, external lab results, or CGM discontinuation. A control group was not assessed for the purposes of this study due to the complexity of retrospectively finding comparable control group patients. Similar future studies should compare pharmacist-led CGM initiation to non-pharmacist-led CGM initiation to more concretely establish the effectiveness of clinical pharmacist interventions. Additional medication management services provided by clinical pharmacists likely further impacted reductions in A1c and may have led to differences seen in our subgroup analysis. Pharmacists may have also seen patients on a more frequent basis, which may have affected the frequency of A1c collection.

Although our analysis demonstrated limited effect on study outcomes from select variables, our ability to analyze all variables was limited by our study's sample size. Our capability to conduct a direct head-to-head analysis between CDTM and MD/NP subgroups was similarly limited by a small sample size in the CDTM arm ($n = 21$). This limits our ability to assess a true comparison between these subgroups.

The limitations of our study stress the need for future prospective trials to assess clinical pharmacist-led CGM initiation with a larger sample size. It may also be worthwhile for these future studies to examine the differences in outcomes

between CGM initiation led by clinical pharmacists vs CGM initiation led by other health care professionals.

Conclusions

To our knowledge, this is the first study that has specifically examined the effect of CGM initiation on A1c reduction among patients with diabetes who were assisted with CGM initiation by a clinical pharmacist. The results of this study demonstrate that CGM device initiation is associated with reductions in A1c and that clinical pharmacists are appropriate health care professionals to assist with this initiation. While our results have demonstrated that these A1c reductions may be even more pronounced among patients followed by clinical pharmacists practicing under CDTM, larger prospective studies are needed to support this assertion.

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Authors' Contribution

James Thurston: Conceptualization, Methodology, Formal analysis, Investigation, Data Curation, Writing – Original Draft, and Visualization. **Hanlin Li:** Conceptualization, Methodology, Writing – Original Draft, Visualization, and Project administration. **Mangala Rajan:** Validation, Formal analysis, Data Curation, Writing – Original Draft, and Visualization. **Yuliya Baratt:** Conceptualization, Methodology, Writing – Original Draft, and Visualization. **Amber Bradley:** Conceptualization, Methodology, Writing – Original Draft, and Visualization. **Fred Pelzman:** Conceptualization, Writing – Review & Editing, and Supervision.

Declaration of Conflicting Interests

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Ethical Statement

Ethical Approval

This protocol was reviewed and approved by the Weill Cornell Medicine Investigational Review Board (IRB) and determined to qualify for exemption from ethical approval.

Informed Consent

This protocol was reviewed and approved by the Weill Cornell Medicine Investigational Review Board (IRB) and determined to qualify for exemption and for waivers of informed consent and HIPAA authorization.

ORCID iDs

James Thurston  <https://orcid.org/0009-0009-6311-6805>

Hanlin Li  <https://orcid.org/0000-0002-6508-145X>

Supplemental Material

Supplemental material for this article is available online.

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